

Assignment–2: OpenMP Performance Evaluation (Exercises)

Name: Rohini Bansal

Platform: Linux (Ubuntu)

Compiler: GCC with OpenMP

1. Objective

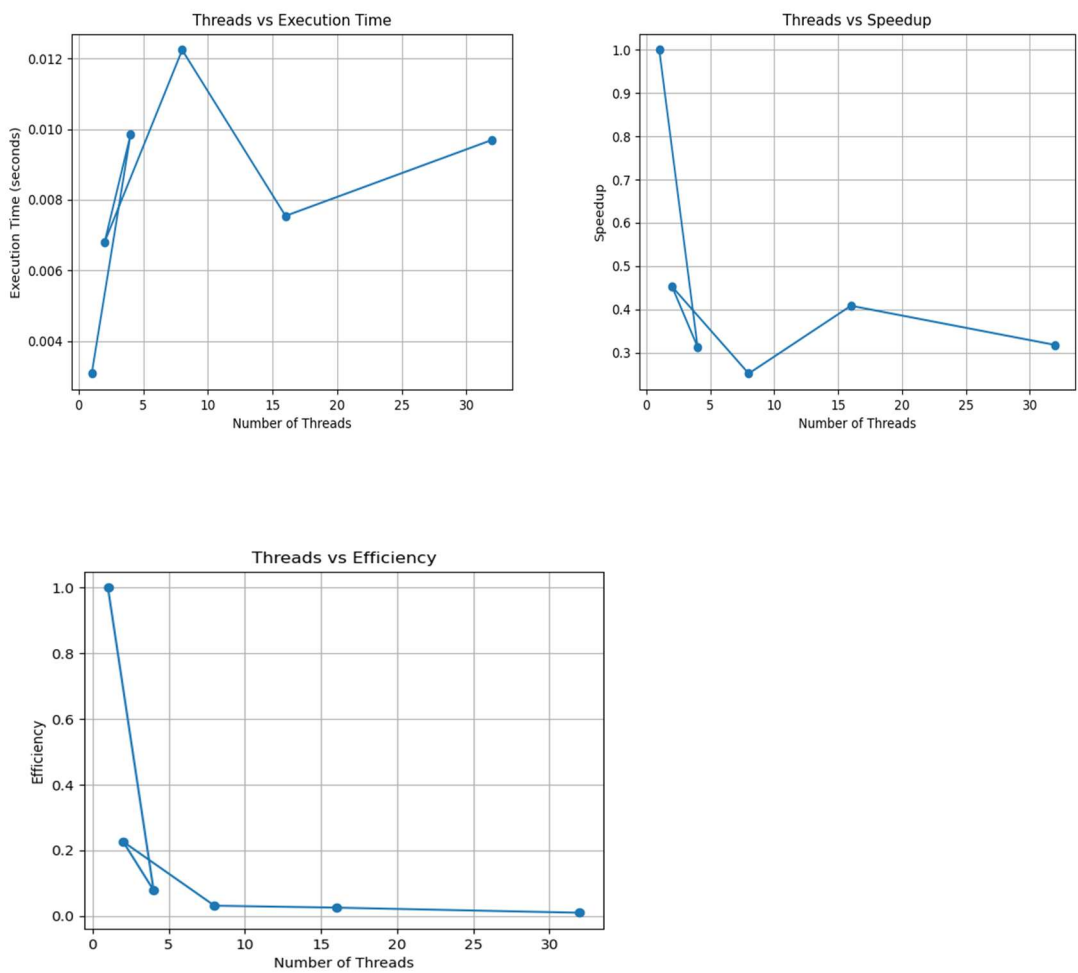
The objective of this report is to implement and evaluate three real-world parallel applications using OpenMP. Performance is evaluated using execution time, speedup, and efficiency metrics, along with scalability graphs.

2. Performance Metrics

Execution Time (T_p) is measured using `omp_get_wtime()`. Speedup $S(p) = T_1 / T_p$ and Efficiency $E(p) = S(p) / p$.

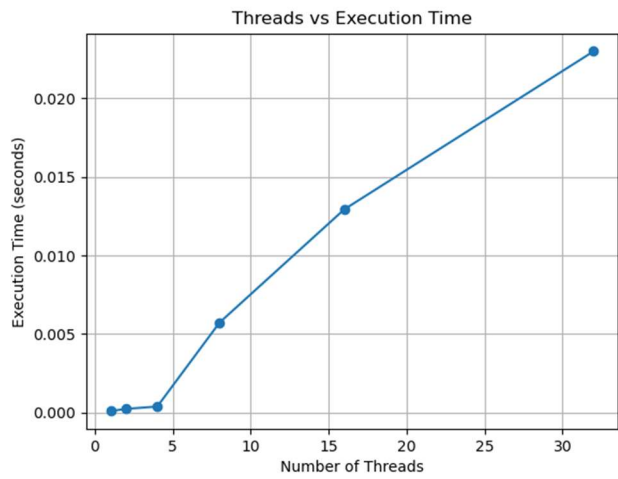
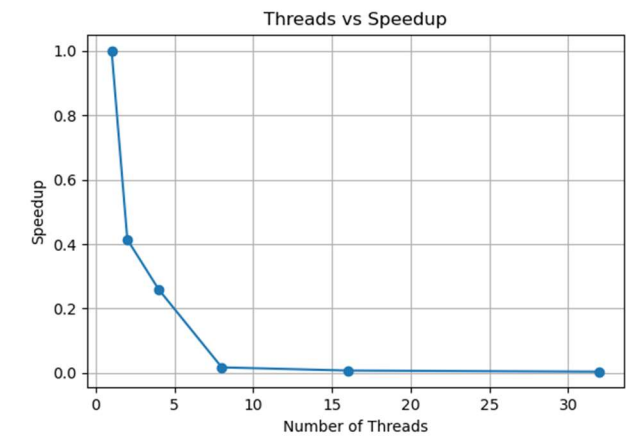
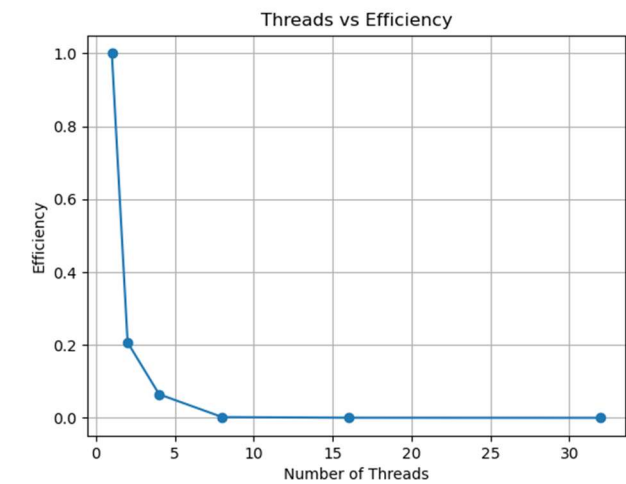
Exercise 1: Molecular Dynamics – Lennard-Jones Force Calculation

The Lennard-Jones potential was used to compute inter-particle forces and potential energy. OpenMP parallelization was applied using dynamic scheduling, atomic operations, and reduction. Execution time was measured to analyze scalability.



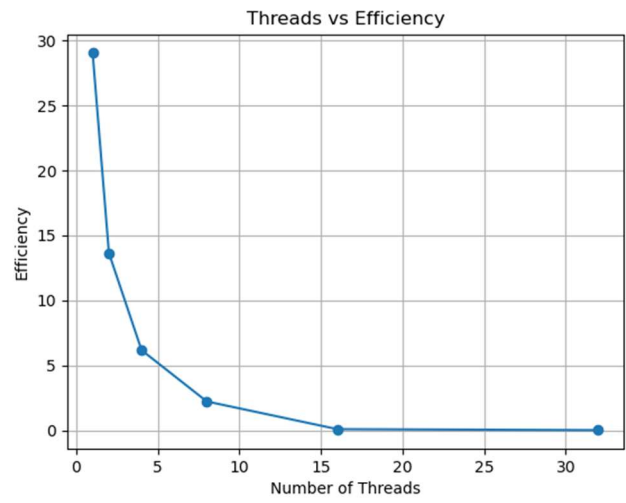
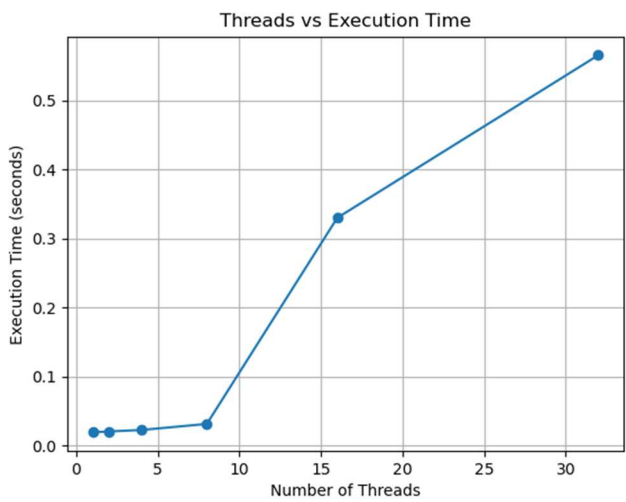
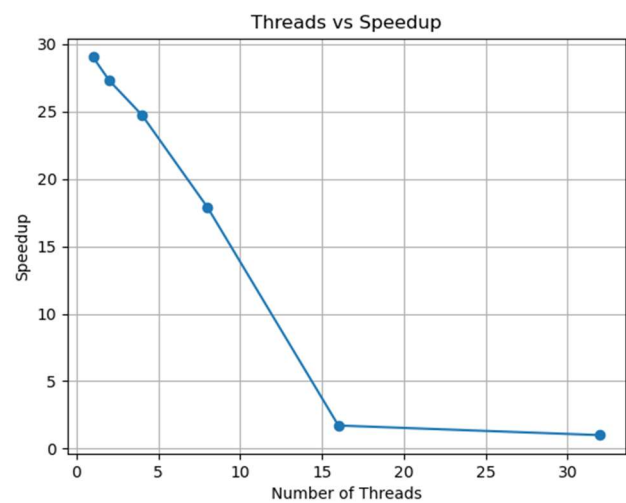
Exercise 2: Bioinformatics – Smith–Waterman DNA Alignment

The Smith–Waterman local sequence alignment algorithm was implemented using wavefront parallelization to handle data dependencies. Performance was evaluated using strong scaling.



Exercise 3: Scientific Computing – Heat Diffusion Simulation

A 2D heat diffusion simulation was implemented using the finite difference method. Spatial loops were parallelized using OpenMP, and performance behavior was analyzed.



Conclusion

All three exercises demonstrate the application of OpenMP performance evaluation techniques. While execution time decreases with increased threads, scalability is limited by synchronization, dependencies, and memory bandwidth.